STABILITY SELECTION IN LOGISTIC REGRESSION

PROJECT REPORT

BY

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DECLARATION

I hereby declare that I am the sole author of this project and that all my work presented is my own, with the exception of references made to the work of other authors. This work has not been submitted as an exercise for a degree to any other university or college. I give Technological University Dublin authority to lend this project report to other institutes or individuals for research purposes.

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ABSTRACT

Variable selection has often presented itself as a challenge in high dimensional statistical problems. The advent and evolution of feature selection methods over the past years has helped resolve this dilemma to some extent, however advanced data storage and modelling techniques tend to introduce new scenarios and unforeseeable settings. Often the robustness of model selection methods is put to test. This report attempts to augment our understanding of the variable selection capabilities of the extant model selection methods in the context of inherent sample variability, extraneous 'interferences' or 'noises' and other design characteristics such as sample size used in the experimental set up. Traditional model selection methods including Likelihood Ratio Test, Stepwise and Lasso regression are first investigated followed by a more recent method, namely the Stability selection method. The entire work is developed around the data provided by the Growing Up in Ireland study, the first national longitudinal study of children in Ireland. The results of the report illustrate a performance comparison between various model selection methods, based on the selection frequency for 'true' predictors and number of 'true' predictors chosen. Amongst the traditional methods, it was noted that Stepwise performed consistently better than LRT and Lasso irrespective of parametric setting considered. Stability selection, on the other hand, performed better overall augmenting the underlying Lasso method by a significant margin in most cases. Amongst some of the demerits observed was the model selection method's propensity to include the unnecessary 'noise' predictors. Stability selection often showed a higher affinity in this regard followed by Stepwise and other methods. Some of the parametric settings also revealed stability selection method's inability to enhance the performance attributes of the underlying methods. This report helps elaborates on these findings in an experimental context, exploring strengths and weaknesses and concludes with two separate discussions into the phenomenon of multicollinearity within 'noise' predictors and the lambda parameter selection within stability selection method. Other possible areas of future research form the final discussion point of this project.

CHAPTER 1
INTRODUCTION

1.1 Purpose of Project

Learning from data has seen a rising impetus in decision making processes. Complex situations, involving uncertainties and randomness, often employs predictive models as an effective information extracting tool. Statistical theory uses a plethora of parametric and non-parametric estimation methods to construct mathematical models capable of such predictive and explanatory power (Larson, 1963). A frequently applied statistical paradigm is one that connects a response variable with a set of decision variables. In statistical model building, it is often acknowledged that the developed model is as good as the predictive power its explanatory variables generates. However, what remains relatively unknown is the exact number of such explanatory variables to be added to the system. This presents to us a situation similar to the 'incompletely specified linear regression problem', coined by Larson (1963). Such class of problems, typically solicitated the use of a selection algorithm that judiciously picks predictor variables with most significant influence on the response variable.

A priori theoretical and practical knowledge should be the primary aid in ascertaining the relevancy of variables. However such information isn't always guaranteed to be at hand or without any unjustifiable bias (Steyerberg, 2009). A formulaic approach is thus recommended in solving such problems. Sequential model building methods witnessed substantial growth and demand by the late 1960's. The fundamental objective was the identification of a right set of variables and their implied functional relationship (Larson, 1963). Many methods posited a criteria-based selection heuristic for choosing the best model from a set of candidate models (Ding, Tarokh and Yang, 2018). However, such a formulation presented a contradicting pair of goals; building a parsimonious model to reduce the overall response variance while maintaining a comprehensive, predictive model encompassing all available information. The incompatibility of these factors called for an appropriate trade off (Geman, Bienenstock and Doursat, 1992). Variable selection methods such as Forward, Backward and their combination, Stepwise, helped resolve this issue to a certain extent (Efroymson, 1960). The introduction of regularization techniques such as Ridge and Lasso gave way to the beginning of

high-performance selection methods. A more recent method involving stability criterion aims to enhance existing selection methods through the introduction of a novel concept called stability paths (Meinshausen and B¨uhlmann, 2009).

Variable selection methods and its performance metrics have been the subject of study in many research works. The unified objective was to gain a better understanding on the efficiency of selection algorithms and infer any conceivable relationship between a selection method and data structure. In medical research, Genell et al. (2010) performed a simulation analysis to contrast Stepwise regression and Bayesian Model Averaging. Suguna et al. (2019) conducted a performance comparison study between various machine learning classification algorithms on Student Academic Performance data from the Kaggle repository. In a more advanced study, Sanchez-Pinto et al. (2018) attempted to differentiate regression based variable selection (including Lasso, Elastic and Stepwise) and Tree based variable selection mechanisms (including Random Forest, gradient boosted feature selection etc.) on the basis of efficiency metrics. Desboulets (2018) demonstrated how 39 variable selection methods, differing in topology, functioned across a common data set. In similar lines, our project aims to deliver a comparative performance study on some variable selection methods including the basic Likelihood Ratio test, Stepwise regression, Lasso and the more recent Stability selection.

1.2 Growing Up in Ireland Study & Data

The data used for this work was collected from the Growing Up in Ireland study (GUI), a longitudinal national study of the youth in Ireland. Our project employed a subset of this data consisting of approximately 6359 responses gathered over 99 different interview-based questions (Smyth, 2017).

Established in 2006 as a government funded initiative, the GUI study was carried out as a collaborative work between researchers from the Trinity College Dublin & the Economic & Social Research Institute (ESRI) to understand the factors affecting children welfare. The study involved findings collated from the same children at different times ; the ‘Child Cohort’ who were first interviewed in 2007 at the age of 9 and at later ages of 13, 17, 18 & 20 years as well as the ‘Infant

Cohort' who were approached first at 9 months of age and followed up after at the ages of 3, 5, 7, 8 & 9 (Smyth, 2017).

In the wake of an increasing number of children retrogressing their first year of secondary education, there was a critical need to understand the transition experiences of children moving to Secondary Schools (McMahon and O'Brien, 2017). One objective of the Growing up in Ireland study was precisely to unearth the fundamental influences surrounding this phenomenon. Smyth (2017), in her study of these transition experiences, employed a multilevel logistic regression concept, to model two types of outcome variables; transition difficulty & school engagement. The multilevel modelling approach helped ensure possible observational data correlations could be accommodated since the design of the study involved both data from sampled clusters like schools and data from the same set of individuals over different times (longitudinal data). A vast set of external factors involving measures of school children's lives and lifestyle were considered and analyzed. Significant primary school engagement experiences were further controlled for.

Amongst the different explanatory factors available, the study chose to focus on how social relationships, primary and secondary school experiences influenced young students' ease of transition. The multilevel model regression output raised several interesting points. Amongst the several social relationship variables, interactions with teachers, friendships, parent's help with homework were judged to be more significant in measuring transition experiences of children. On similar lines, student's attitude towards school, subjects like English, Irish & Math were adjudged significant primary-school related variables. A student being in the first or second year of secondary schools was also found to have considerable influence in explaining transition difficulties.

This project uses the R statistical software to access the GUI data and extract regression results. A logistic regression model is fitted using the above variables, to offer preliminary insights into the explanatory power generated. The same model is further used as a standard, to understand and compare the variable selection capability of certain methods discussed later in this report.

1.3 Logistic Regression

The outcome variables considered by the GUI study were mostly binary in nature, with values of either a 1 (success) or a 0 (failure). The distribution of the response data, thus, represented a binomial distribution.

The probability mass function of a binomial random variable is given by:

$$\Pr(Y = y) = \binom{n}{y} p_i{}^y \, (1 - p_i)^{n-y} \, , y = 0,1,2,3 \ldots n \qquad (1.1)$$

Where Y is the number of successes in n trials & p_i is the probability of success of each binary random variable

Fitting a linear statistical model to such binomial or categorical response variables, however, presented a challenge. Regressing probabilistic parameters over a set of explanatory variables, introduced sigmoidal shaped response variations, hinting at the non-normal distributions of the error as well as the limited range of the categorical parameter. A suitable transformation of nonlinear regression models, using link functions, helps carry over linear regression techniques to categorical variables (Nelder and Wedderburn, 1972). Nalder and Wedderburn (1972)'s work titled Generalized Linear Models conceptualizes the above methodology to a wider class of regression estimation problems involving distributions from the general exponential family of distributions.

The multi variable GLM regression equation for a binomial response variable Y_i of the ith sample is given by:

$$g(\mu_i) = X_i'\beta \qquad (1.2)$$

where $g(*)$ is the link function, μ_i is the mean of the binomial response variable Y_i, $X'\beta$ is the linear predictor in matrix notation; X_i is the corresponding design matrix & β the unknown coefficient vector

Y_i follows any distribution from the general exponential distribution density function given as follows:

$$f(y, \theta, \psi) = \exp\left\{\frac{y\theta - b(\theta)}{a(\psi)} + c(y, \psi)\right\} \quad (1.3)$$

where $a(*), b(*), c(*)$ are functions varying with the type of distribution of Y_i while θ, ψ are two parameters of the distribution

Logistic regression is a GLM that helps connect a linear predictor model with the binomial response variables using the logit (link) function, shown below:

$$\text{logit}(p_i) = \log\left(\frac{p_i}{1 - p_i}\right) \quad (1.4)$$

where p_i is the probability of succeses of binomial response variable

A complete logistic linear model is represented as,

$$\text{logit}(p_i) = \log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 x_{1i} + \ldots\ldots + \beta_k x_{ki} \quad (1.5)$$

where $x_{1i}, x_{2i}, \ldots . x_{ki}$ are predictor variables and $\beta_0, \beta_1, \ldots \beta_k$ are coefficients of the unknown predictor variable coefficients

Alternatively, the probability or proportion of successes of the binomial response can be expressed as,

$$p_i = \frac{\exp(\beta_0 + \beta_1 x_{1i} + \ldots\ldots + \beta_k x_{ki})}{1 + \exp(\beta_0 + \beta_1 x_{1i} + \ldots\ldots + \beta_k x_{ki})} = \frac{\exp(X_i'\beta)}{1 + \exp(X_i'\beta)} = \frac{\exp(\eta_i)}{1 + \exp(\eta_i)} \quad (1.6)$$

where η_i is the linear predictor

To build a logistic model, estimating the transition experiences described in the GUI, the report's multimodal regression results were extensively studied. A list of possible response-predictor variables was extracted and evaluated. A secondary school student's engagement with mathematics, adequately represented a transition experience and hence was chosen as the response variable. The regressors selected, included transition difficulty, student's attitude towards primary school and primary level mathematics subject, measure of positive or negative interaction with teachers, gender of a student, student's current year in secondary school & the highest qualification a secondary

student seeks to attain in the future. Taking into account the categorical nature of the predictors, indicator variables were introduced to consider individual levels of the factors. Overall, 24 such indicator variables were considered while setting up the design matrix of the predictor coefficients. The rank deficiency and over parametrization of this matrix was rectified using a set to zero constraint resulting in a final design matrix with 18 indicator variables (Searle, 1987).

The design matrix X, containing all 18 indicator variables for 8 predictors across 6359 observations is shown below,

$$X = \begin{bmatrix} 1 & x_{1,1,1} & x_{1,1,2} & \cdots & x_{1,8,n} \\ 1 & x_{2,1,1} & x_{2,1,2} & \cdots & x_{2,8,n} \\ 1 & x_{3,1,1} & x_{3,1,2} & \cdots & x_{3,8,n} \\ & \vdots & \vdots & \cdots & \vdots \\ 1 & x_{6359,1,2} & \cdots & \ldots & x_{6359,8,n} \end{bmatrix} \quad (1.7)$$

where $x_{a,b,c}$ is the indicator variable for the bth predictor's c^{th} level in a^{th} observation

while coefficient vector β is given by,

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \\ \beta_5 \\ \beta_6 \\ \vdots \\ \beta_{18} \end{bmatrix} \quad (1.8)$$

where β_i is the coefficient of the ith indicator variable

The final step in statistical modelling, is determining a set of model implied parameters that forms a statistically significant regression equation. Amongst different methods available for computing the estimates, the Maximum Likelihood Estimation helps produce the desirable unbiased estimates (Guo, 2013). It involves maximizing the likelihood of reproducing a sample output, i.e. the observed probability of a response predictor model.

Maximizing the likelihood function proceeds by evaluating the joint probability distribution of N samples' response variables as shown below,

$$f(y) = \prod_{i=1}^{N} \binom{n_i}{y_i} p_i^{y_i} (1 - p_i)^{n_i - y_i} \tag{1.9}$$

To estimate the predictor coefficients β, p_i is substituted with the values derived in Eq 1.6, to arrive at the likelihood function L(β) given below.

$$L(\beta) = \prod_{i=1}^{N} \binom{n_i}{y_i} \left(\frac{\exp(X'\beta)}{1 + \exp(X'\beta)}\right)^{y_i} \left(\frac{1}{1 + \exp(X'\beta)}\right)^{n_i - y_i} \tag{1.10}$$

A log likelihood, $\ell(\beta)$, is often taken for computational convenience resulting in the following,

$$\ell(\beta) = \sum_{i=1}^{N} \ln\left(\binom{n_i}{y_i}\right) + y_i (X'\beta) - n_i \log (1 + \exp(X'\beta)) \tag{1.12}$$

The predictor coefficients β are estimated solving for the below maximization equation,

$$\hat{\beta} = \arg\max_{\beta}\big(\ell(\beta)\big) \tag{1.13}$$

Since the maximization problem does not yield a closed form solution, numerical approximation methods are often implemented. Newton Raphson (N.R) method, an efficient iterative algorithm, gives convincing results (Agresti, 2002).

The method begins with some assumed values for the β matrix. Using the available data inputs from the X matrix, the algorithm churns out an estimate of $\ell(\beta)$. Another iteration with new values for the β matrix, produces a second estimate of $\ell(\beta)$ and a possible direction for subsequent guess values of the β matrix. In midst of multiple β values however, finding a direction can be quite challenging, necessitating the use of higher order partial derivatives of $\ell(\beta)$. The N.R uses two matrices to that effect, a (p×1) vector having the first order partial derivatives of $\ell(\beta)$ called the 'Score Vector' S_j and a (p×p) matrix having second order partial derivatives of $\ell(\beta)$ called the

'Hessian Matrix' H_j. The inverse of the Hessian matrix gives the subsequent direction for guess values of the β matrix, thus helping converge to the final solution.

For the predictor model used in this study, these matrices are represented as,

$$S_j = \begin{bmatrix} \frac{\partial \ln \ell_j}{\partial \beta_0} \\ \frac{\partial \ell_j}{\partial \beta_1} \\ \vdots \\ \frac{\partial \ell_j}{\partial \beta_{18}} \end{bmatrix} \tag{1.14}$$

$$H_j = \begin{bmatrix} \frac{\partial^2 \ln \ell_j}{\partial {\beta_0}^2} & \frac{\partial^2 \ln \ell_j}{\partial \beta_0 \, \partial \beta_1} & \cdots & \frac{\partial^2 \ln \ell_j}{\partial \beta_0 \, \partial \beta_{18}} \\ \frac{\partial^2 \ln \ell_j}{\partial \beta_1 \, \partial \beta_0} & \frac{\partial^2 \ln \ell_j}{\partial {\beta_1}^2} & \cdots & \frac{\partial^2 \ln \ell_j}{\partial \beta_1 \, \partial \beta_{18}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 \ln \ell_j}{\partial \beta_{18} \, \partial \beta_0} & \frac{\partial^2 \ln \ell_j}{\partial \beta_{18} \, \partial \beta_1} & \cdots & \frac{\partial^2 \ln \ell_j}{\partial {\beta_{18}}^2} \end{bmatrix} \tag{1.15}$$

The matrix of β estimates obtained from the process, are finally tested for their statistical significance to complete logistic regression modelling.

In the R software, the cumulation of the steps described above resulted in the following outcomes,

$$\hat{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \\ \beta_5 \\ \beta_6 \\ \vdots \\ \beta_{18} \end{bmatrix} = \begin{bmatrix} -0.6665 \\ 0.18551 \\ 0.22080 \\ -0.0051 \\ 0.40353 \\ 0.03793 \\ 1.04975 \\ 0.46973 \\ -0.4652 \\ -0.3579 \\ -0.2925 \\ -1.0452 \\ -0.6855 \\ -0.3858 \\ -0.3800 \\ -0.2400 \\ -0.2434 \\ 0.05743 \\ -0.0528 \end{bmatrix}$$

Where,

β_0 represents the Intercept

β_1 represents the factor level trans_diff_qrt(13,15]

β_2 represents the factor level trans_diff_qrt(15,35]

β_3 represents the factor level trans_diff_qrt(5,10]

β_4 represents the factor level acce14Never like it

β_5 represents the factor level acce14Sometimes like it

β_6 represents the factor level acce16aNever like it

β_7 represents the factor level acce16aSometimes like it

β_8 represents the factor level tposscale > 3

β_9 represents the factor level tposscale2.75

β_{10} represents the factor level tposscale3

β_{11} represents the factor level tnegscale1

β_{12} represents the factor level tnegscale1.5

β_{13} represents the factor level tnegscale2

β_{14} represents the factor level bcce37Second year

β_{15} represents the factor level bphc02amale

β_{16} represents the factor level bcce50Degree or higher degree

β_{17} represents the factor level bcce50Junior Cert

β_{18} represents the factor level bcce50Leaving Cert

The values obtained above corresponds to the predictor coefficients of all the variables (including levels) of the predictor mode. β_1 to β_3 represent different levels of the transition difficulty variable, β_4 to β_5 represent different levels of the variable representing the student's attitude towards primary school, β_6 to β_7 represent different levels of the variable for a student's attitude towards primary level mathematics subject, β_8 to β_{10} represent different levels of the variable measuring the student's positive interaction with teacher, β_{11} to β_{13} represent different levels of the variable measuring the student's negative interaction with teacher, β_{14} represent a level of the variable referring to the student's current year in secondary school, β_{15} represent a level of the variable indicating the gender of a student and β_{16} to β_{18} represent different levels of the variable suggesting the highest qualification that a secondary student seeks to attain in the future. Amongst these coefficients, β_0, β_2, β_4, β_6, β_7, β_8, β_9, β_{10}, β_{11}, β_{12}, β_{13}, β_{14}, β_{15} & β_{16} were concluded statistically significant based on R's summary statistics. The numerical values of these coefficients can be interpreted as the magnitude of influence of each predictor, on the probability of success of the response variable (student's engagement with mathematics at secondary school). Since β_6 has the highest value among the list of predictor coefficient estimates, it can be inferred that student's attitude towards mathematics at primary level, plays a significant role in determining his/her level of engagement for the same subject at secondary level.

These regression coefficients will be further used in simulation studies, to evaluate the performance of model selection methods, described in the next chapter of this report.

CHAPTER 2
MODEL BUILDING METHODS

2.1 Introduction

Our data repository from the Growing Up in Ireland study has close to a hundred variables (Smyth, 2017). Some of these variables may have explanatory power while others are just pure noise. Identifying the 'right' set of predictors from such an extensive data, becomes a challenging task considering that it involves competing notions of creating an all-inclusive (more informative, incorporating every possibly related parameter) that would benefit from being simultaneously sparse (less complex). Some of the earliest known model building strategies performed exhaustive searches across all the possible candidate models. The underlying idea was to add all variables simultaneously in a model. The best subset algorithm (Kutner et al., 2005) is one consideration, where the objective is to parse and select the best fitting model, from a subset of all possible models, using a selection criterion such as the Wald's Test, Likelihood Ratio Test or Score Test. However, this method was often dataset-dependent, rendering its results extremely variable (Breiman, 1996). Additionally, the models generated became increasingly complex with the number of predictors, thus calling for further research in this area. Sequential addition of variables started gaining attention by early with automated search procedures like Stepwise selection and its variants (forward and backward selection) producing simpler and more consistent models. However, such methods were still not without the risk of developing an inflated Type 1 error owing to the multiple tests conducted for the inclusion or exclusion of each variable. Appropriate corrections help mitigate these concerns but only to a certain extent (Miller, 1981). Consequently, several measures, such as Mallow's Cp, Akaike Information Criteria (AIC) and its variant Bayesian Information Criteria (BIC), were introduced as the better criteria for assessing the goodness of fit of the model. However, this introduced yet another problem; overfit models. A breakthrough for mitigating this issue came with the realization of how multicollinearity of predictors can be the cause for instability & possible overfitting of the model. This further led to the introduction of orthogonal transformation procedures such as principal component analysis and later on, penalized regression (regularization) methods. The Ridge Regression (Hoerl and Kennard, 1970), followed by the Lasso algorithm (Tibshirani, 1996) were among the first model selection procedures to employ regularization parameters within Ordinary Least Squares (OLS) formulation. The contention was

that the regularization parameter would ensure a price for increasing the value of the regression coefficients, effectively shrinking any coefficient perceived as noise. This implies that the estimated coefficients would not be overly reliant on the underlying data, hence decreasing the chance for overfitting.

In this study, three kinds of model selection methods are primarily investigated; a non-sequential model selection using the Likelihood Ratio Test, sequential Stepwise algorithm, Lasso method and Stability selection method.

2.1 Likelihood Ratio Test

The Likelihood Ratio Test (LRT) testing criterion, developed by Neyman and Pearson (1928), conceptualized likelihood measure, as a parametric standard for comparing the significance of fitted covariates. From a model building perspective, LRT implies selecting a subset of the initially fitted base model. This is done via a multiple comparison test, testing for the statistical significance of each predictor fitted in the base model.

From the context of hypothesis testing, every comparison made, would have a Null Hypothesis H_0, assuming a predictor coefficient from the base model to be of zero value, competing with an Alternate Hypothesis H_a, assuming otherwise. H_a, thus maps to the base model while H_0, maps to a reduced model having one predictor less than the base model. The decision test criterion, computed as twice the ratio of log likelihoods or difference in deviance between the two models, is compared against a standard test criterion. The asymptotic distribution of the decision test statistic under H_0 usually follows a Chi Squared distribution. Comparing this with a standard Chi Squared Statistic having degrees of freedom equal to the difference in number of predictors between the models, estimates the statistical significance of the extra predictor present in the base model.

$$\begin{aligned} \text{LRT} &= -2(\ln(\mathcal{L}_s) - \ln(\mathcal{L}_b)) \\ &= -2\ln(\mathcal{L}_s) + 2\ln(\mathcal{L}_b) \\ &= \text{deviance}_s - \text{deviance}_b \end{aligned} \tag{2.1}$$

where $\mathcal{L}_s$ is the likelihood of the reduced model and $\mathcal{L}_b$ is the likelihood of the base model

In the R statistical software, the LRT is implemented using the drop1 command.

2.2 Stepwise Model Selection

Stepwise regression, an explanatory data analysis technique introduced by Efromyson (1960), identifies from a predictor pool, the correct subset of independent variables to be added to a model (Mc Intyre et al., 1983). As an automated sequential algorithm, it examines the statistical significance of adding new predictors or removing fitted ones, in single 'steps' and subsequently implements one of the actions. Identifying an appropriate criterion for additions or deletions has been a subject of debate, since they are known to be quite specific to the type of regression encountered. For example, David Hosmer (2000) noted how model selection techniques using an F test criterion worked differently on a linear gaussian model compared to a logistic regression one; F Test criterion applied on normally distributed random variations gave more convincing results. Another argument raised was that, certain criteria, at times produced erroneous parametrical inferences, resulting in parameter biases, overfit models & incorrect significance tests (Whittingham et al., 2006). This ultimately led to the development of an improved criterion, the Akaike Information Criteria (AIC), which has ever since, seen substantial application in information theoretic model selection. The AIC is represented as,

$$\mathrm{AIC} = -2\ln(\mathcal{L}_{\mathrm{m}}) + 2(\mathrm{p}) \qquad (2.2)$$

where $\mathcal{L}_{\mathrm{m}}$ *is the likelihood of the model being tested &* p *is the number of model parameters*

Though structurally similar to the LRT statistic, the AIC additionally penalizes for increasing the number of parameter estimates, hence keeping a check on model overfit. The lower the AIC, the better the model. A variant, the Bayesian Information Criteria (BIC), is sometimes preferred, in order to introduce higher parametrical penalization. Choosing between either of these criteria is more, a matter of context.

Stepwise variable selection for the GUI logistic regression model, using an AIC criterion, is performed using the following steps (Hosmer and Lemeshow, 2000):

1. An intercept only (baseline) model is first fitted to the GUI dataset. This model's AIC is computed using the relation in equation 2.1.

2. Every predictor from the predictor pool is evaluated for the change in AIC it brings, when added to the model. If the current AIC value decreases as a consequence of the inclusion, the new predictor is considered worthy of addition.

3. At any stage, when at least on predictor has been added to the model, each predictor in the model is again evaluated for the change in AIC if removed from the model. If the AIC value decreases as a consequence of the removal, the considered predictor is considered worthy of deletion.

4. The process stops when neither adding nor removing a predictor reduces the AIC any further or when the predictor that is to be removed was the predictor, added in the previous step.

2.3 LASSO Model Selection

Sequential model building methods using 'improved' selection criteria (AIC or Mallows Cp), were not entirely successful in precluding 'noise' (irrelevant) predictors from entering the model and thus, occasionally, reverted to the overfit models and erroneous conclusions described in the previous section. The results were often unstable, implying that the variables selected varied depending on the structure of the underlying dataset. Another criticism raised was that such methods offered no protection against inflation of regression coefficients.

Penalized regression methods such as Ridge Regression (Hoerl and Kennard, 1970) & Least Absolute Shrinkage and Selection Operator (LASSO) (Tibshirani, 1996), attempted to resolve some of the shortcomings of the sequential building methods. It introduced the concept of 'shrinkage of parameters' in order to optimize the bias-variance trade off and improve the predictability of the model. Depending on the penalization parameter chosen, the perceived 'noise' parameters are shrunken to zero values, thus improving the variable selection.

In a simple linear regression problem, the predictor coefficient $\hat{\beta}_\lambda$ is obtained by solving for the minimization of an OLS ℓ_1 regularization model with respect to β, shown below:

$$\hat{\beta}_\lambda = \arg\min_\beta \left(\|Y - X\beta\|_2^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right) \quad (2.3)$$

where Y is a continous normally distributed variable, X is the design matrix for regression problem, $\sum |\ |$ denotes the ℓ_1 norm , λ is the regularization parameter & β_0, $\beta_1, \ldots \beta_j$ are regression coefficients of the predictor variables

The concept is almost similar in a logistic regression model, the only difference being that the OLS is replaced by the Likelihood Function discussed in section 1.3. The coefficients in this case are obtained by minimizing the resulting penalized log likelihood function given below,

$$\hat{\beta}_\lambda = \arg\min_\beta \left(-\ell(\beta) + \lambda \sum_{j=1}^{p} |\beta_j| \right) \quad (2.4)$$

The above ℓ_1 penalized log likelihood function can be minimized using appropriate algorithms such as the full gradient algorithm (Goeman, 2010) or the split and conquer method (Shofiyah and Sofro, 2018). A suitable regularization parameter, λ is vital for the quick convergence of the Lasso estimator as well as the risk consistency of the resulting predictor coefficients (Chatterjee, 2014). Risk consistency, as a measure relates to the prediction risk of the Lasso estimator. Hence, as the λ parameter increases, it becomes more expensive to simultaneously increase the β_j's, resulting in their diminution. Greenshtein and Ritov (2004) demonstrated that a K- fold Cross Validation Approach for estimating the λ parameter yielded positive results with respect to the conditions required for risk consistency.

A K- Fold Cross Validation technique for estimating λ parameter in a penalized logistic regression makes use of the log likelihood function $\ell(\beta)$ and is performed with the following steps:

1. The dataset is partitioned into k equally sized segments or folds. The model is fitted (trained) using k-1 folds and validated using the unused kth fold.

2. A range of λ values are considered starting from 0 and including λ_{max} (which shrinks all predictors to 0) as shown in Fig 2.1. For each λ value, the $\hat{\beta}_\lambda$ are calculated using the data from the training set (k-1 folds), solving for the below relation.

$$\hat{\beta}_{-i,\lambda} = \arg\min_\beta(-\ell_{-i}(\beta_{-i,\lambda}) + \lambda \sum_{j=1}^{p}|\beta_j|) \quad (2.5)$$

where $\ell_{-i}(\beta_{-i,\lambda})$ is the penalized loglikelihood using the training dataset (excluding the ith fold)

The $\hat{\beta}_\lambda$'s obtained are then used to find deviance of the validation set (kth fold), using the equation below.

$$\ell(\hat{\beta}_{-i,\lambda}) = \sum_{i=1}^{N} \ln\left(\binom{n_i}{y_i}\right) + y_i\left(X'\hat{\beta}_\lambda\right) - n_i \log\left(1 + \exp(X'\hat{\beta}_\lambda)\right) + \lambda \sum_{j=1}^{p}|\hat{\beta}_{\lambda_j}| \quad (2.6)$$

where $\ell(\hat{\beta}_{-i,\lambda})$ is the deviance measure calculated on the validation set

3. The mean deviance on the validation set is then computed using the following relation.

$$\text{Mean Deviance} = \frac{1}{k}\sum_{i=1}^{k}\ell(\hat{\beta}_{-i,\lambda}) \quad (2.7)$$

4. Step 2 & 3 are repeated for different values of λ and the λ with the highest value for Mean Deviance is finally selected.

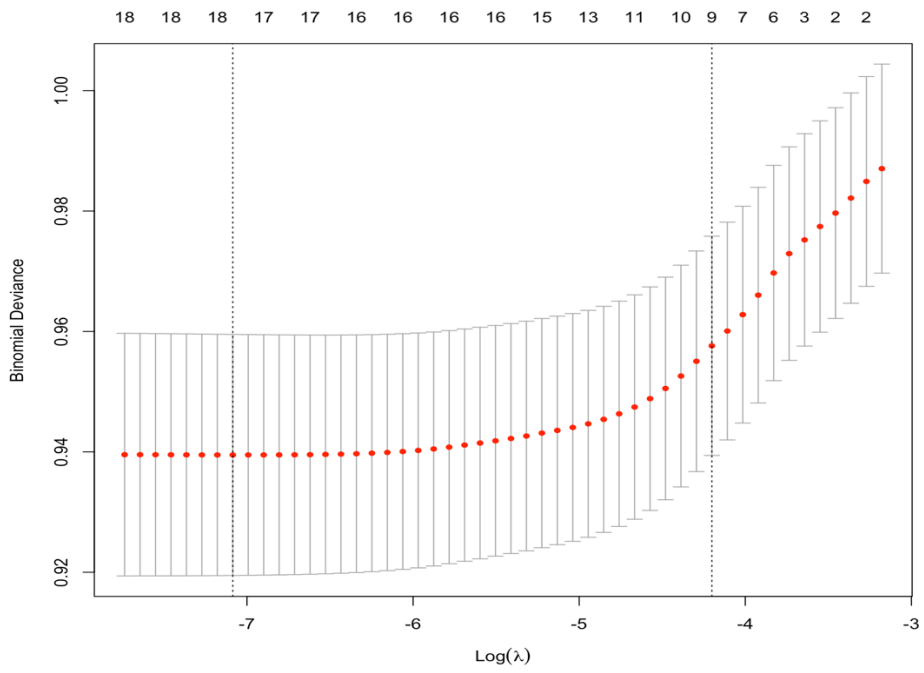


Figure 2.1: Cross validation plot showing a Lambda that shrinks all coefficients to zero

To derive the Lasso estimator of the model, recall the logistic regression equations 1.5 and 1.6 discussed in section 1.3. The Bernoulli Distribution of the response variable is given by,

$$f(y|\beta) = \binom{n_i}{y_i} p_i^{y_i} (1-p_i)^{n_i - y_i} \tag{2.8}$$

where y_i is the binary response variable we are modelling having values of 0 or 1

The likelihood and the log likelihood are calculated using the joint probability distribution,

$$L(\beta) = \prod_{i=1}^{n} \binom{n_i}{y_i} p_i^{y_i} (1-p_i)^{1-y_i} \tag{2.9}$$

$$\ell(\beta) = \ln\left(\sum_{i=1}^{n} \binom{n_i}{y_i} p_i^{y_i} (1-p_i)^{1-y_i}\right)$$

$$= \ln\left(\sum_{i=1}^{n} \binom{n_i}{y_i} p_i^{y_i} (1-p_i)^{1-y_i}\right)$$

$$= \sum_{i=1}^{n} \ln\left(\binom{n_i}{y_i}\right) + y_i \ln(p_i) + (1 - y_i)\ln(1 - p_i)$$

$$= \sum_{i=1}^{n} \ln\left(\binom{n_i}{y_i}\right) + y_i \ln\left(\frac{p_i}{1 - p_i}\right) + \ln(1 - p_i)$$

$$= \sum_{i=1}^{n} \ln\left(\binom{n_i}{y_i}\right) + y_i\eta_i - \ln(1 + \exp(\eta_i))$$

The lasso estimator for the logistic regression model is hence given by,

$$\beta_{\hat{\lambda}} = \min_{\beta}\left(\sum_{i=1}^{n} -\ln\left(\binom{n_i}{y_i}\right) - y_i\eta_i + \ln(1 + \exp(\eta_i))\right) + \lambda \sum_{j=1}^{p} |\beta_j| \qquad (2.10)$$

2.4 Stability Selection

Reproducibility is a universal, scientific concept regarded essential for the effective implementation of statistical investigations. In feature selection literature, reproducibility is often proxied by the term 'stability' and refers to the robustness of the selected features to random variations in data (Křížek, Kittler and Hlaváč, 2007). Empirical studies however reveal that despite its relevance, stability is a parameter that is usually met unsatisfactorily by many model selection algorithms. Experimental results often confirmed this inference. Morozova et al. (2015) demonstrated how subset selection methods such as Stepwise & Lasso regression produced inconsistent results across epidemiological datasets. Works of Henkal, Martin and Nardari (2011) displayed similar outcomes in time series models when return predictions of stock market revealed empirical evidence, disproving stable predictive power over time series data. Collective analysis of all such results revealed multicollinearity as the common culprit. In case of Stepwise and Lasso regression, their inability to handle predictor-collinearity, indirectly invalidates the stability criterion, since a different predictor is selected, from a common pool of correlated predictors, for different datasets (Nogueira and Brown, 2016). In response to this issue, Zou and Hastie (2003) introduced the Elastic net Lasso, an improved version of Lasso regression, using a convex combination of ℓ_1 and ℓ_2 regularization. Doing so, offered this technique offered a competitive edge; retaining the sparsity of ℓ_1 regularization all the while including the grouping effect of ℓ_2 regularization (Nogueira and Brown, 2016). This grouping effect ensured that correlated predictors receive the same coefficient

value including the possibility of a sign reversal, in cases of negative correlation, thus averting the predicament seen in Lasso & Stepwise regression. However, this method necessitated the estimation of a suitable regularization parameter λ for optimal performance.

Meinshausen and B¨uhlmann (2009) circumvented the problem of regularization through a subsampling technique. Amongst other benefits, the subsampling approach essentially helped induce consistent variable selection and thus controlled Familywise type 1 Error or the false discovery rate of relevant parameters in finite samples (Meinshausen and B¨uhlmann, 2009). The underlying intuition behind Stability Selection algorithm was that a sequence of data perturbations followed by feature selection, when repeated enough number of times created probabilistic patterns for assessing the selection rate of a feature (Křížek, Kittler and Hlaváč, 2007). The predictors selected, are further deemed stable if their overall frequency of selection, is greater than a particular threshold. From the stability point of view, repeated or iterative sampling, potentially produced varying datasets or conditions, which provided insights into the efficacy of a variable selection method. The selection probability of individual predictors essentially measured this.

This report utilizes the concept of stability selection applied to a Lasso logistic regression model as outlined in section 2.3. Stability selection is performed with the following steps,

1. A bootstrap sample with 5000 entries is first generated from the GUI dataset. Cross validation is performed on this sample to obtain a range of λ's for the GUI logistic regression model. λ's between the minimum and one standard deviation away from the minimum, are considered for inclusion and stored in a set L.

2. From the sample in step 1, a second sample is generated without replacement and with size around half of that in step 1.

3. A Lasso model is fitted to the sample obtained in step 2 using each of the candidate λ's in L. The selected predictors are noted.

4. Steps 2 & 3 are iterated 500 times to note the number of times a predictor was selected for a particular λ from set L. This is labeled as the selection probability of this predictor.

5. The maximum value of this selection probability is determined over the range of λ values available. If the value is greater than a predefined minimum threshold (0.7 as per this study), the predictor is considered selected. The final model will reflect all predictors selected in this manner.

Having explored the different kinds of model selection methods, the subsequent step in this research is to examine their working and mechanism under different scenarios. The next chapter introduces the setting up of a simulation methodology, required to replicate commonly observed data patterns and evaluate the model selection methods considered.

CHAPTER 3
SIMULATION STUDY

3.1 Simulated Model Selection

Evaluating performance characteristics using simulation methods has been a much-researched topic within statistical communities. Morris, White and Crowther (2019), described how simulation studies helped measure different performance measures based on the type of statistical tasks. For an 'estimation' task, the evaluated performance metrics include bias, consistency and standard error of the estimate. A 'testing' task, on the other hand, measured the Type 1 error rate associated with a Null Hypothesis, while a third category, 'model selection', involved measures such as the correct model selection rate or the specificity for a predictor selection (Morris, White and Crowther, 2019).

In the 'model selection' world, with the range of metrics available, selecting an appropriate performance measure boils down to the type of problem at hand. Regression problems often use model's efficacy for covariate selection, as a performance standard. This metric becomes more significant in high dimensional settings, where the performance of models is greatly limited by the presence of noise patterns. The complication mostly observed in such cases was the correlation of noise variables with some of the relevant features, allowing for their 'piggybacked' entry into the model (Flack and Chang, 2012). 'Noise' predictors, in this context refers to regressors that do not contribute to the estimation of a regressed variable. In a broader sense however, it can include random variations following any kind of distribution.

3.2 Design parameters of the Simulated Model Selection

Against the background of the logistic model developed in section 1.2, the following design parameters are set for the simulation study.

a) Performance Measures

The primary performance measures explored through this simulation are the selection frequency for each predictor and the selection frequency of the different numbers of predictors selected.

b) 'True' & 'Noise' predictors

All variables included in the GUI logistic regression model developed in section 1.2, are considered 'true' predictors. 'Noise' parameters are further added to the model to test the model selection method's robustness against random variations. In this study, 'noise' variations are introduced via two methods; by adding predictors irrelevant to the response variable or by the adding a random variate to the linear predictor of the model.

The randomly chosen 'noise' predictors are of two kinds; predictors which are related to response due to multicollinearity alone and the predictors that are truly 'noise'. In this work, the 'noise' predictors include are variables associated with adequacy of number of teachers in secondary school, parent's attendance in Parent-Teacher's meeting, primary caretaker's highest qualification, family's social background, religious ethos of school & availability of an induction day at school. The normal random variates, acting as the second type of 'noise' variable, are generated using a mean of 0 and standard deviation of 4 units.

c) Sample size

The simulation was performed using two different samples, of size 5000 and 500. The idea was to understand the role of sample size in the simulation outcomes.

d) Model Selection Methods

Variable selection methods including LRT, Stepwise, Lasso and Stability selection were evaluated as a part of this simulation set up.

e) Scenarios

The following scenarios are explored in this simulation.

1) Model selection with 'true' predictors alone
2) Model selection with 'true' and 'noise' predictors
3) Model selection with 'true' predictors and random 'noise' in the linear predictor.

3.3 Pseudo Code for the Simulated Model Selection (LRT & Stepwise)

The below pseudo code is utilized for performing the LRT & Stepwise model selection simulation.

1. Fit a regression model to the given data set using a suitable response variable and some 'true' predictors.
2. Define set P of all 'true' (i^{th}) predictors (p_i) and set N of all 'noise' (i^{th}) predictors (n_i)*.
3. Define and initialize a set of counters C_i, for each (i^{th}) predictor that a model selection algorithm may select.
4. Start model selection simulation for x runs
    i. For each run d
        1. Generate a bootstrap sample of size n from the dataset.
        2. Add a normal random variate* to the linear predictor developed in step 1 and predict response values from this predictor.
        3. Randomize the above values.
        4. Perform LRT (Stepwise) model selection using 5 % significance level deviance criterion (AIC) criterion and define set S_d accumulating each (j^{th}) predictor variable(s) ($s_{d,j}$) selected in the dth run*.
            a. If ($s_{d,j} \in P$)
                i. Increment C_i for the corresponding p_i.
            b. Else if ($s_{d,j} \in N$)
                i. Increment C_i for the corresponding n_i.
            c. End of if.
    ii. End of for loop for simulation run

5. End of model selection simulation.

Note * denotes that the step is conditional on the scenario being taken for the simulation study.

3.4 Pseudo Code for the Simulated Model Selection (Lasso)

The below pseudo code is utilized for performing the Lasso model selection simulation.

1. Fit a regression model to the given data set using a suitable response variable and some 'true' predictors.
2. Define set P of all 'true' (i^{th}) predictors (p_i) and set N of all 'noise' (i^{th}) predictors (n_i)*.
3. Define and initialize a set of counters C_i, for each (i^{th}) predictor that a model selection algorithm may select.
4. Start model selection simulation for x runs
    i. For each run d
        1. Generate a bootstrap sample of size n from the dataset.
        2. Add a normal random variate* to the linear predictor developed in step 1 and predict response values from this predictor.
        3. Randomize the above values.
        4. Using a 10-fold cross validation find λ_{min} value that maximizes the mean deviance of the validation dataset fitted with the above randomized response values.
        5. Perform model selection using a $\lambda_{1-S.E}$ value, one standard deviation away from λ_{min} and define set S_d accumulating each (j^{th}) predictor variable(s) ($s_{d,j}$) selected in the dth run*.
            a. If ($s_{d,j} \in P$)
                i. Increment C_i for the corresponding p_i.
            b. Else if ($s_{d,j} \in N$)
                i. Increment C_i for the corresponding n_i.
            c. End of if.
    ii. End of for loop for simulation run

5. End of model selection simulation.

Note * denotes that the step is conditional on the scenario being taken for the simulation study.

3.5 Pseudo Code for the Simulated Model Selection (Stability Selection)

Development of a pseudo code for stability selection reused many of the steps described for the Lasso algorithm. The additional steps required, accounted for the subsampling approach used by stability selection. The pseudo code is illustrated below.

1. Fit a regression model to the given data set using a suitable response variable and some 'true' predictors.
2. Define set P of all 'true' (i^{th}) predictors (p_i) and set N of all 'noise' (i^{th}) predictors (n_i)*.
3. Define and initialize a set of counters C_i, for each (i^{th}) predictor that the main model selection algorithm may select.
4. Start main model selection simulation for x runs
    i. For each run d
        1. Define and initialize another set of counters H_i, for each (i^{th}) predictor that a model selection method may select within subsamples. Also define a common threshold value T for each H_i that deems the respective predictor selected.
        2. Generate a sample of size n.
        3. Add a normal random variate* to the linear predictor developed in step 1 and predict response values from this predictor.
        4. Randomize the above values
        5. Using a 10-fold cross validation find λ_{min}value that maximizes the mean deviance of the validation dataset fitted with the above randomized response values.
        6. Consider a set λ_d of λ values from λ_{min} to $\lambda_{1-S.E}$ (one standard deviation away from λ_{min}) .**
        7. Start subsample model selection simulation for y runs.
            a. For each run e

i. Generate a bootstrap sample of size n/2 from the dataset.
ii. Add a normal random variate* to the linear predictor developed in step 1 and predict response values from this predictor.
iii. Randomize the above values.
iv. Perform model selection using each (j^{th}) $\lambda_{d,j}$values generated in the dth run of step 5 and define set $S_{d,\lambda_{d,j}}$ accumulating each (k^{th}) predictor variable(s) ($s_{e,\lambda_{d,j},k}$) selected in the eth run (the current loop).
1. If ($s_{e,\lambda_{d,j},k} \in P$)
a. Increment H_i for the corresponding p_i.
2. Else if ($s_{e,\lambda_{d,j},k} \in N$)*
a. Increment H_i for the corresponding n_i.
3. End of if.
b. End of for loop for sub sample model selection simulation run.
8. End of sub sample model selection simulation.
9. If ($H_i >= T$)
a. Corresponding predictor p_i or n_i is considered selected; Increment C_i for $p_i$or n_i.*
10. End of if.
ii. End of for loop for main model selection simulation run.
5. End of main model selection simulation

Note: * denotes that the step is conditional on the scenario being taken for the simulation study.
** denotes that the step is conditional on the type of model selection method the stability selection is applied to.

CHAPTER 4
RESULTS OF MODEL SIMULATION

4.1 Introduction

The performance measures, selection frequency of true or 'noise' predictors and the selection frequency of different number of predictors selected, were evaluated against the simulation design parameters discussed in section 3.2. The experimental data results obtained, required further examination and analysis to bring out the right insights. This section explains the methodologies used in this study, to help assist with this process. A logistic regression model was developed, to delineate the impact of simulation study design parameters on the probability of selecting any predictor. For selection frequencies of number of predictors selected, appropriate histogram plots of the selection counts data helped visualize the probabilistic distributional patterns. The statistical significance of any distributional difference observed, was further established using the Kruskal Wallis test and a post hoc analysis, Wilcoxon signed rank test.

4.2 Impact of simulation parameters on selection frequency of predictors

The effects of design parameters, on selection frequencies of individual predictors, were studied using a logistic regression model. A binary indicator variable, representing the selection of a predictor, was regressed across 3 factor variables; model selection methods, predictors and type of sample used in the experiment. As per the design parameters, a 'method' variable, representing the different model selection methods, would have 4 different levels, corresponding to each variable selection method, while the 'type' factor variable would have 8 factor levels for 'true' predictors and 6 such levels for 'noise' predictors. A 'sample' factor variable with 2 levels would account for the size of the sample used i.e. 5000 or 500.

In logistic terms, the estimates obtained from the regression model, represent the log odds ratio for a unit increase in the corresponding predictor variable values as discussed in section 1.3. Adding up any set of coefficient estimates including the intercept, provides a quantitative measure of the influence exerted by each design parameter on the response variable. For example, consider a simple three predictor logistic model shown below,

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

Here, the coefficient β_1 measures the increase in log odds of the response, per unit increase in the corresponding exposure i.e. variable x_1. Similarly, the sum of β_1 and β_2 indicates the combined effect of both variables $x_1$and x_2. A log-odds ratio greater than 1 indicates an exposure variable associated with higher odds of outcome while a ratio less than 1 signifies lower odds of outcome.

Using the above logic, we consider 3 different logistic models to account for the different scenarios mentioned in section 3.2. The first model investigates the effect of a chosen sample's random variations in altering the selection rate of predictors. This is the base model, which considers 'true' predictors alone. A second model evaluates the impact of adding some 'noise' predictors to the base model while the third and final one looks into the effect of random normal 'noise' variables (Normal R.V).

The models and their effects are described in detail below.

(a) Model 1- 'True' Predictors alone

In implementing this scenario, all the variables other than the 'type' variable is retained in the model. A total of 15 coefficients were initially accounted for, however, considering over parametrization of the design matrix this was reduced to 12 coefficients, with the effects of any dropped coefficient being taken care of by the intercept term. An implementation of the logistic model in R, yielded the following results for the values and statistical significance of the coefficient estimates.

$$\hat{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \\ \beta_5 \\ \beta_6 \\ \beta_7 \\ \beta_8 \\ \beta_9 \\ \beta_{10} \\ \beta_{11} \end{bmatrix} = \begin{bmatrix} 1.78884 \\ -3.1395 \\ -2.3696 \\ -2.6399 \\ -3.1609 \\ -0.2566 \\ -1.6988 \\ -2.8938 \\ 4.77593 \\ -0.8124 \\ 1.08892 \\ 3.23093 \end{bmatrix}$$

Where,

β_0 represents the Intercept

β_1 represents the ′type′ variable′s factor level corresponding to acce14 predictor

β_2 represents the ′type′ variable′s factor level corresponding to bhpc02a predictor

β_3 represents the ′type′ variable′s factor level corresponding to bcce50a predictor

β_4 represents the ′type′ variable′s factor level corresponding to bcce37 predictor

β_5 represents the ′type′variable′s factor level corresponding to tnegscale predictor

β_6 represents the ′type′ variable′s factor level corresponding to tposscale predictor

β_7 represents the ′type′ variable′s factor level corresponding to trans_diff_qrt predictor

β_8 represents the ′method′ variable′s factor level corresponding to Lasso model selection

β_9 represents the ′method′ variable′s factor level corresponding to LRT model selection

β_{10} represents the ′method′ variable′s factor level corresponding to Stability selection

β_{11} represents the ′sample′ variable′s factor level corresponding to a 5000 sample size

All coefficients estimates were found to be statistically significant.

As mentioned in section, the 'true' predictors represented here include those variables indicating a student's attitude towards primary school (β_1), gender of student (β_2), highest qualification that a secondary student seeks to attain in the future (β_3), student's current year in secondary school (β_4), student's negative and positive interaction with teacher (β_5 and β_6) as well as transition difficulties of student (β_7).

Inspection of the coefficients and probability values showed that predictors related to both student's current year in secondary school and their attitude towards primary school had the least chances for being selected, irrespective of the model selection method chosen. On the other hand, variables influencing a student's negative and positive interaction with teachers, were highly likely, to be selected by any of the methods considered. Amongst the variable selection methods, Lasso showed higher affinity for selecting 'true' predictors; it selected all such predictors at least 90% of the times. Stability selection applied to Lasso failed to improve Lasso and reduced the selection probability by different amounts across all predictors, the lowest being 42%. The negative coefficient of LRT equated to a meagre 15% selection probability for some predictors (e.g. student's current year in secondary school) and higher probability (of more than 50%) for others (e.g. student's negative interaction with teachers). Overall, this meant that, the LRT was more likely to miss out choosing the right predictor variables.

(b) Model 2- Adding 'Noise' Predictors ('True' and 'Noise' Predictors)

Adding 'noise' predictors to the previously mentioned model introduces 6 new variables. A logistic regression produced the following set of coefficients values,

$$\hat{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \\ \beta_5 \\ \beta_6 \\ \beta_7 \\ \beta_8 \\ \beta_9 \\ \beta_{10} \\ \beta_{11} \\ \beta_{12} \\ \beta_{13} \\ \beta_{14} \\ \beta_{15} \\ \beta_{16} \\ \beta_{17} \end{bmatrix} = \begin{bmatrix} -3.1728 \\ 0.09333 \\ 0.00415 \\ -0.6015 \\ 4.39652 \\ 1.75284 \\ 2.33286 \\ 2.36454 \\ -0.7760 \\ 1.50016 \\ -0.7733 \\ 4.10562 \\ 3.04510 \\ 1.86534 \\ -1.3090 \\ -0.6360 \\ 2.40975 \\ 2.81403 \end{bmatrix}$$

Where,

β_0 represents the Intercept

β_1 represents the 'type' variable's factor level corresponding to bpsd40a predictor

β_2 represents the 'type' variable's factor level corresponding to bdsd57a predictor

β_3 represents the 'type' variable's factor level corresponding to bhpc61a predictor

β_4 represents the 'type' variable's factor level corresponding to acce16a predictor

β_5 represents the 'type' variable's factor level corresponding to acce14 predictor

β_6 represents the 'type' variable's factor level corresponding to bhpc02a predictor

β_7 represents the 'type' variable's factor level corresponding to bcce50a predictor

β_8 represents the 'type' variable's factor level corresponding to bhpc55 predictor

β_9 represents the 'type' variable's factor level corresponding to bcce37 predictor

β_{10} represents the 'type' variable's factor level corresponding to bpce33a predictor

β_{11} represents the 'type' variable's factor level corresponding to tnegscale predictor

β_{12} represents the 'type' variable's factor level corresponding to tposscale predictor

β_{13} represents the 'type' variable's factor level corresponding to trans_diff_qrt predictor

β_{14} represents the 'method' variable's factor level corresponding to Lasso model selection

β_{15} represents the 'method' variable's factor level corresponding to LRT model selection

β_{16} represents the 'method' variable's factor level corresponding to Stability selection

β_{17} represents the 'sample' variable's factor level corresponding to a 5000 sample size

The values above, represent the coefficient estimates for all the predictors; caretaker's highest qualification (β_1), family's social background (β_2), availability of an induction day at school (β_3), student's attitude towards primary level mathematics (β_4), student's attitude towards primary school (β_5), gender of student (β_6), highest qualification that a secondary student seeks to attain in the future (β_7), religious ethos of school (β_8), student's current year in secondary school (β_9), parent's attendance in Parent-Teacher's meeting (β_{10}), student's negative and positive interaction with teacher (β_{11} and β_{12}) as well as transition difficulties of student (β_{13}).

Comparing the different coefficient values, it is apparent that there was some level of consistency in picking up variables related to student's aptitude for primary level mathematics and their positive or

negative interaction with teachers. A similar analysis for the effects of different model selection methods, showed that Stability selection performed the best amongst all methods considered. It produced a high 85% probability of selection, nearly two to three times higher than that of any other methods. The fact that Lasso, in the absence of stability selection, performed with a much lower selection capacity (15%), demonstrates the performance enhancing capability of the Stability selection for the case presented.

Computing the selection frequencies of 'noise' predictors revealed a startling observation. Stability selection, given its heightened proclivity to include the right predictors, oversteps and includes the non-essential 'noise' predictors as well, during the course of variable selection. Lasso, in comparison, does a better job in precluding such noise variables. Another observation noted was that ranking order for 'noise' predictor selection did not differ from that for the true predictors, implying that all model selection methods were equally likely to select either a 'noise' or true predictor.

(c) Model 3- Effect of inclusion of 'Noise' variates– ('True' predictors and Normal R.V)

Adding normal random variates as 'noise' to the model 1, resulted in the following set of coefficient estimates.

$$\hat{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \\ \beta_5 \\ \beta_6 \\ \beta_7 \\ \beta_8 \\ \beta_9 \\ \beta_{10} \\ \beta_{11} \end{bmatrix} = \begin{bmatrix} -0.6612 \\ -2.1682 \\ -1.3501 \\ -1.6757 \\ -2.2279 \\ -0.1364 \\ -1.0426 \\ -2.0024 \\ -1.9524 \\ -0.8596 \\ 0.81794 \\ 2.94931 \end{bmatrix}$$

Where,

β_0 represents the Intercept

β_1 represents the $'$type$'$ variable$'$s factor level corresponding to acce14 predictor

β_2 represents the ′type′ variable′s factor level corresponding to bhpc02a predictor
β_3 represents the ′type′ variable′s factor level corresponding to bcce50a predictor
β_4 represents the ′type′ variable′s factor level corresponding to bcce37 predictor
β_5 represents the ′type′variable′s factor level corresponding to tnegscale predictor
β_6 represents the ′type′ variable′s factor level corresponding to tposscale predictor
β_7 represents the ′type′ variable′s factor level corresponding to trans_diff_qrt predictor
β_8 represents the ′method′ variable′s factor level corresponding to Lasso model selection
β_9 represents the ′method′ variable′s factor level corresponding to LRT model selection
β_{10} represents the ′method′ variable′s factor level corresponding to Stability selection
β_{11} represents the ′sample′ variable′s factor level corresponding to a 5000 sample size

A striking difference noted in this case was the all-positive numeric values for the logistic regression coefficients. The obvious implication was a drastic reduction in the selection probability of predictors, compared to previous scenarios. The positive - negative interaction variables were most frequently selected followed by the variable relating to a student's gender. In contrast, student's current year in secondary school variable was almost negligibly chosen, with probability estimates often less than 15%. Comparing the coefficients for selection methods, Lasso was the least resistant to 'noise' disruption, as it rarely picked up any of the 'true' predictor over the course of iterations. Stability applied to Lasso, offered some improvement by raising Lasso's selection probability (26.3%), however this value was still much lower than those obtained in any of the other scenarios considered so far. Similar outcomes were noted for the remaining selection methods as well.

(c) Effects of sample size

A recurring observation, during the course of the experiment, was the large coefficient values of the '5000' sample size variable. The natural implication was that the size of sample used had a massive influence on the selection frequencies (probabilities) of predictors. The probabilistic variations were further found to be invariant across scenarios and methods explored in this study.

4.3 Effect of simulation parameters on selection frequency of number of predictors

The precision, with which a variable selection algorithm selected different sets of 'true' predictors, showed a spectrum of results when plotted with a histogram. Most of these histograms were in fact skewed implying a non-normal distribution of the number of predictors selected. Thus, usual parametric tests like ANOVA failed to give conclusive results, in discerning differences between the distributions. With the assumptions of normality broken, it is more practical to resort to alternate nonparametric versions.

The non-parametric Kruskal Wallis test, when employed, helps correctly evaluate the obtained distributions. A standard Kruskal Wallis test is described below.

Kruskal Wallis Test

One of the earliest nonparametric tests to have been developed for non-normal distribution involved the use of ranked data (Wilcoxon, 1945). Though these methods were initially considered for one or two sample test, the technique was later extended by Kruskal and Wallis (1952) to cover multiple samples analogous to the ANOVA parametric test widely used today. The concept of ranked data also meant that the parameter under consideration would be median as opposed to mean, which is more commonly used in parametric versions.

This study makes use of the Kruskal Wallis (KW) test to understand the statistical significance of any perceived difference between the distributions of number of 'true' and 'noise' predictors selected across each model selection method. The hypothesis statements are set up as below:

$$H_0 = None\ of\ the\ distribution\ for\ the\ counts\ of\ different\ number\ of\ 'true'$$

$$('noise')\ predictors\ selected\ stochastically\ dominated\ one\ other\ distribution.$$

$$H_a = Atleast\ one\ of\ the\ distribution\ for\ the\ counts\ of\ different\ number\ of\ 'true'$$

$('noise')\ predictors\ stochastically\ dominated\ one\ other\ distribution.$

The statistic used for the test is a Chi squared distributed H statistic,

$$H = \left[\frac{12}{n(n+1)}\sum_{j=1}^{s}\frac{R^2{}_j}{n_j}\right] - 3(n+1) \qquad (4.1)$$

$Where,$

$n\ is\ the\ total\ number\ of\ datapoints$

$s\ is\ the\ \ number\ of\ groups\ being\ compared$

$R_j is\ the\ sum\ of\ the\ ranks\ in\ the\ j^{th} group$

$n_j\ is\ the\ number\ of\ datapoints\ within\ each\ group$

The standard statistic, against which the H statistic is compared, is the critical Chi Squared value at 5% significance level with s – 1 degrees of freedom.

The Kruskal Wallis test proceeds as follows:

1. The dataset comprising the frequency distribution of the different number of selected predictors is sorted in the ascending order.

2. The individual datapoints are ranked using numerical values in the same ascending order. Any tied data are given an average rank based on the number of such points.

3. The ranks are added up within each model selection method group.

4. The H statistic is computed using the above values and compared with the critical value. If the p value is less than the 5 % significance level, the Null hypothesis is rejected.

Rejection of the null hypothesis from the KW test, calls for a second examination to identify the groups that are different from one another. For such multiple comparisons, a post hoc test called the Wilcoxon signed rank test is utilized. A Holm-Bonferroni method of correction further ensures that inflated type 1 and type 2 error rates introduced via multiple comparisons are checked and maintained within appropriate ranges.

The effects of the design parameters on the model selection simulation of a 500-size sample are examined first, followed by a comparison with the results for the 5000-size version.

(a) 'True' Predictors alone

In the case, where only 'true' predictors were added to the logistic model, the model selection algorithms produced histogram plots shown in Figure 4.1.

The plots indicated that the Stability selection applied to Lasso, didn't introduce a discernable difference in Lasso's performance. Numerical evidence proved that both accuracy (selecting the correct number of 'true' predictors) and precision (consistently selecting a number of predictors) fell by a small margin, upon introducing the stability algorithm to Lasso. Nevertheless, stability selection managed to grab 6 predictors as a median value with the actual number of predictors selected ranging mostly between 4 and 6 predictors as seen in Figure 4.1. In the order of performance, Lasso was adjudged the best followed closely by Stability selection, Stepwise and LRT.

To test for the significance of these observed differences, a Kruskal Wallis test is performed. An implementation in the R software gave a p value of less than the 5 % significance level indicating that at least one of the distributions was different from the others. A Wilcoxon signed rank test was further conducted to examine the differences across all pairwise comparisons. This test too returned p values less than the 5% significance level value, suggesting that there was enough evidence in the distribution patterns to conclude statistically, that any two pairs of distribution pattern of the model selection method were not the same.

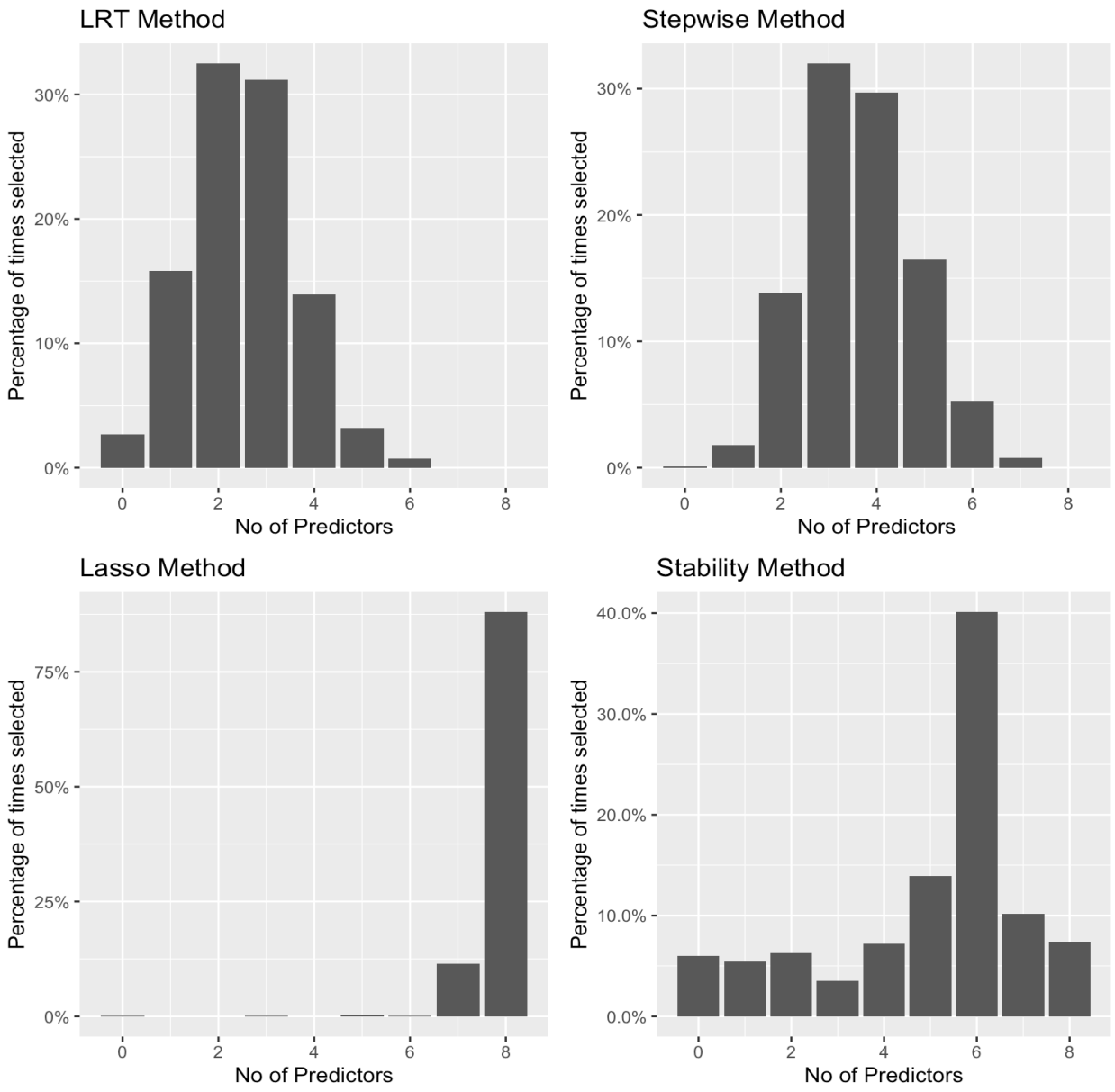


Figure 4.1 Histogram plots for number of 'true' predictors selected across model selection methods with no other effects considered.

The final conclusion drawn, from the results of the statistical tests, was that all the methods presented and tested in this case scenario selects 'true' predictors in different capacities and patterns, with the Lasso showing the best, near to optimal, performance while LRT towards the bottom end.

(b) Effect of inclusion of 'Noise' Predictors ('True' and 'Noise' Predictors)

Compared to the case with 'true' predictors alone, the addition of 'Noise' predictors managed to diminish the number of predictors selected by many methods as seen from Figure 4.2.

Lasso's performance dropped considerably with the median reaching the zero mark, although the Stability selection still retained the number of selected predictors at 6. One of the other influences of the 'noise' predictor was to increase the range of the number of predictors selected. The only exception to this was the Stepwise method, where there wasn't a noticeable change in the distributional pattern as can be seen comparing Figure 4.1 with Figure 4.2.

A Kruskal Wallis Chi squared value of 2028.7, implied a difference between at least one pair of the distributions observed. The results from the Wilcoxon test, further indicated that all the distributions were statistically different from one another.

A similar plot for the 'noise' predictors selected, showed that on average all the methods other than Stability selection, were less prone to select any more than a single 'noise' predictor. In this case Stability selection also showed a larger range for the number of 'noise' variables selected. Lasso performed the best in terms of keeping out the 'noise' with good accuracy and precision, followed closely by the LRT and Stepwise as shown in Figure 4.3.

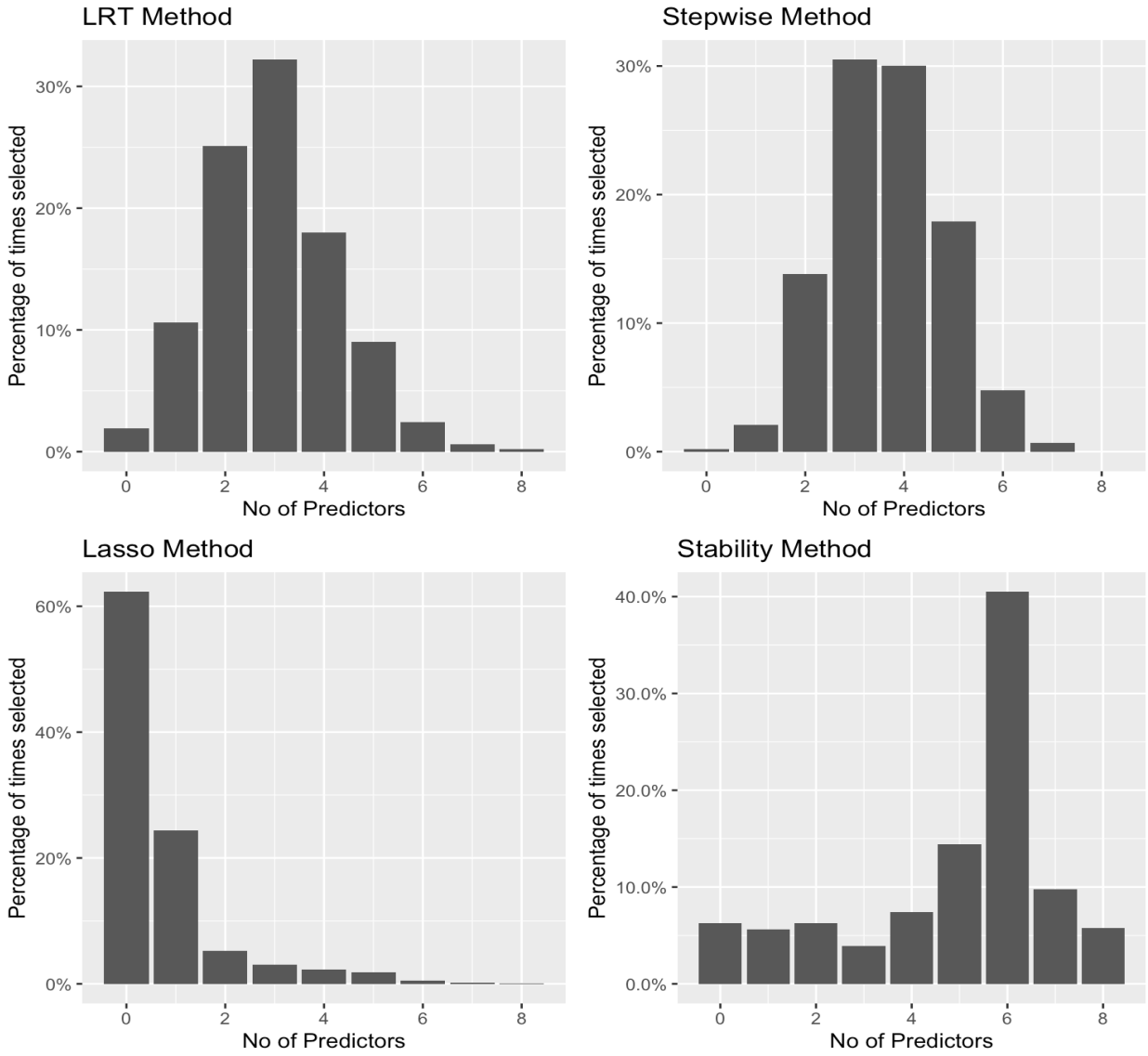


Figure 4.2 Histogram plots for the number of 'true' predictors selected across model selection methods in the presence of 'noise' predictors

The KW test and the Wilcoxon test both showed that these observed differences between the distributions of the model selection methods were statistically significant.

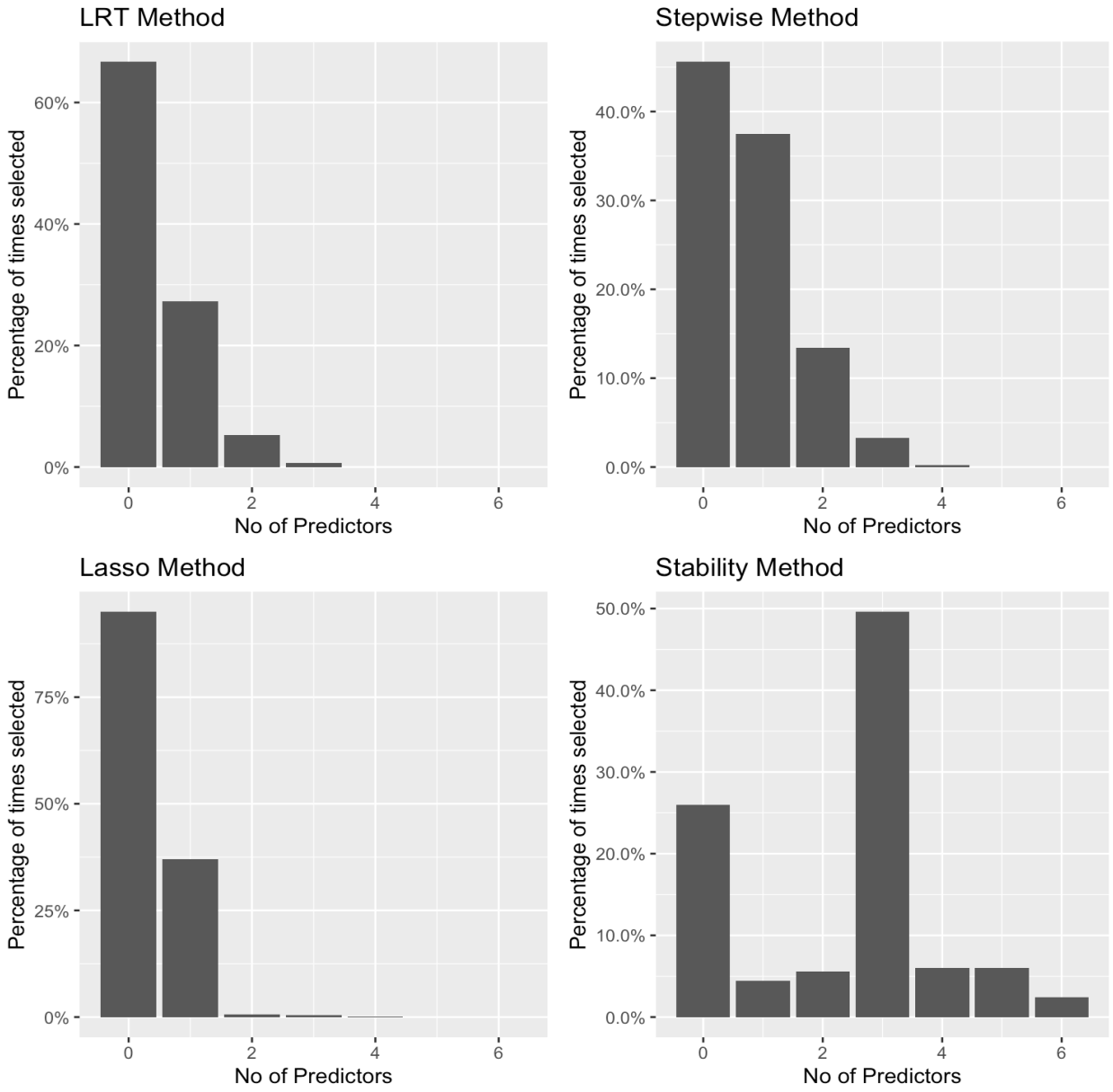


Figure 4.3 Histogram plots for the number of 'noise' predictors selected across model selection methods in the presence of 'noise' predictors

(c) Effect of inclusion of 'Noise' variates– ('True' predictors and Normal R.V)

A larger margin of difference was observed when random normal variates where added to the system.

In contrast to the cases discussed so far, most methods, limiting the Stepwise to some extent, showed a higher probability to miss out selecting any of the right predictors, as seen in Figure 4.4.

The peaks in the histogram plots were concentrated around the 0-4 predictors range, with the only exception noted for stability selection, which was almost equally likely to select either 6 or none of the predictors.

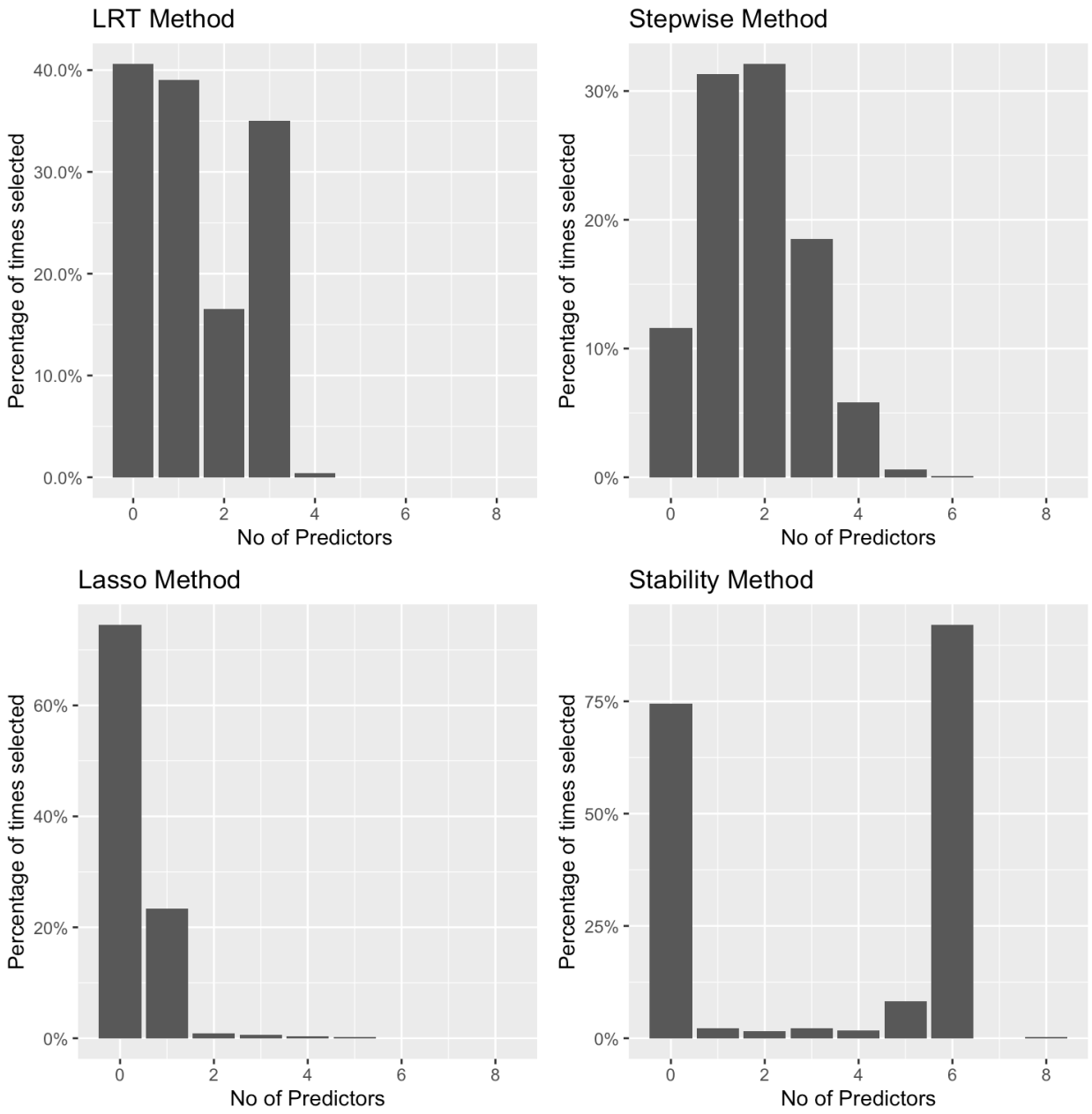


Figure 4.4 Histogram plots for the number of 'true' predictors selected across model selection methods in the presence of random normal variables

The overall spread of the selection range was drastically reduced when compared to the previous scenarios. When examined with a KW test and the Wilcoxon test there was enough evidence to believe that all model selection methods selection frequency distributions were statistically different thus confirming all observations raised above.

(c) Effect of sample size

Similar to the effect of sample size described in section 4.1, increasing the number of observations in a sample to 5000 had the potent effect of increasing the number of predictors selected irrespective of the method chosen. With respect to distributional differences, in the cases involving 'true' predictors alone, KW and Wilcoxon test pointed out statistical evidence proving the similarity of selection frequencies for Stability selection and Lasso method. A similar observation was also noted for the distributions of LRT and Lasso in the cases with 'noise' predictors.

All the outcomes discussed in this section of the experimental study are summarized in the tables 4.5(a) and 4.5(b). The tables show performance measures 'Selection Frequency of Predictors' and 'Selection Frequency of Number of Predictors' across different scenarios and sample sizes introduced in this chapter. The primary numeric values displayed in the table, relates to the two performance measures discussed in section 3.2 of this report i.e. the selection probability for all the 'true' predictors and the median scores of the number of such predictors selected by the different variable selection methods. An additional interquartile range for the number of predictors selected gives a sense of the number of predictors that are selected more frequently. Calibrating and analyzing these metrics, in essence, help understand and compare the performance variations across different variable selection methods.

Sample Size	"Noise"	Scenario	Performance Measure	Result
500	No	'True' predictors alone	Selection Frequency of Predictors (Probability of selecting 'True' Predictors)	Lasso > Stability > Stepwise > LRT (98.3%) (65.5%) (44.2%) (29.2%)
			Selection Frequency of Number of predictors (Median score of the number of 'True' Predictors selected) & (Interquartile Spread)	Lasso > Stability > Stepwise > LRT (8) (6) (4) (2) -- 4 -- 6 3 -- 4 2 -- 3
	Yes	+ 'Noise' predictors	Selection Frequency of Predictors (Probability of selecting 'True' Predictors)	Stability > Stepwise > LRT > Lasso (84.8%) (38.8%) (26.4%) (16.4%)
			Selection Frequency of Number of predictors (Median score of the number of 'True' Predictors selected) & (Interquartile Spread)	Stability > Stepwise > LRT > Lasso (6) (4) (3) (0) 4 -- 6 3 -- 4 2 -- 4 0 -- 1
			Selection Frequency of 'Noise' Predictors (Probability of selecting 'Noise' Predictors)	Stability > Stepwise > LRT > Lasso (26.6%) (3.9%) (2.2%) (1.25%)
			Selection Frequency of Number of 'noise' predictors (Median score of the number of 'Noise' Predictors selected) & (Interquartile Spread)	Stability > Stepwise > LRT > Lasso (3) (1) (0) (0) 0 -- 3 0 -- 1 0 -- 1 --
		+ 'Noise' Normal R.V	Selection Frequency of Predictors (Probability of selecting 'True' Predictors)	Stability > Stepwise > LRT > Lasso (26.3%) (14.5%) (7.3%) (3%)
			Selection Frequency of Number of predictors (Median score of the number of 'True' Predictors selected) & (Interquartile Spread)	Stepwise > LRT > Stability = Lasso (2) (1) (0) (0) 1 -- 3 0 -- 1 0 -- 1 0 -- 1

Table 4.5 (a) Summary of results obtained for 500 sample size

Sample Size	"Noise"	Scenario	Performance Measure	Result
5000	No	'True' predictors alone	Selection Frequency of Predictors (Probability of selecting 'True' Predictors)	Lasso > Stability > Stepwise > LRT (99.9%) (96.7%) (92.5%) (86.6%)
			Selection Frequency of Number of predictors (Median score of the number of 'True' Predictors selected) & (Interquartile Spread)	Stability = Lasso > Stepwise = LRT (8) (8) (7) (7) -- -- 7 -- 8 6 -- 7
	Yes	+ 'Noise' predictors	Selection Frequency of Predictors (Probability of selecting 'True' Predictors)	Stability > Stepwise > LRT > Lasso (98.5%) (88.3%) (81%) (70%)
			Selection Frequency of Number of predictors (Median score of the number of 'True' Predictors selected) & (Interquartile Spread)	Stability > Stepwise = Lasso = LRT (8) (7) (7) (7) -- 7 -- 8 6 -- 8 6 -- 7
			Selection Frequency of 'Noise' Predictors (Probability of selecting 'Noise' Predictors)	Stability > Stepwise > LRT > Lasso (81.1%) (32.8%) (21.6%) (13.1%)
			Selection Frequency of Number of 'noise' predictors (Median score of the number of 'Noise' Predictors selected) & (Interquartile Spread)	Stability > Stepwise > Lasso = LRT (6) (1) (0) (0) -- 0 -- 1 0 -- 1 0 -- 1
		+ 'Noise' Normal R.V	Selection Frequency of Predictors (Probability of selecting 'True' Predictors)	Stability > Stepwise > LRT > Lasso (83%) (70%) (51.6%) (29.3%)
			Selection Frequency of Number of predictors (Median score of the number of 'True' Predictors selected) & (Interquartile Spread)	Stability > Stepwise > LRT > Lasso (8) (5) (4) (2) 7 -- 8 4 -- 6 3 -- 5 0 -- 4

Table 4.5 (b) Summary of results obtained for 5000 sample size

CHAPTER 5
CONCLUSION & SYNTHESIS

5.1 Summary

The discussions in the previous chapter helped generate a great deal of contextual perspectives on various model building method's performance characteristics. The most general inference drawn was that the results obtained, provides reasonable evidence to suggest performance enhancement capabilities of stability selection, when wrapped around the Lasso model. Selection precision and selection accuracy were the performance measures that saw a marked improvement. This effectively translated to the selection method capturing the right predictors repeatedly apart from selecting a greater number of the true predictors just as often as well. However, the only contradicting point would be the scenario dealing with randomizations or the inherent variability within samples. Stability selection offers no improvement here and rather deteriorates Lasso's overall performance both in accuracy and precision. Amongst the traditional methods, there doesn't seem to be a single answer to which model selection method can be adjudged best. Lasso does exceptionally well against inherent randomizations (variability) but unlike Stepwise or LRT doesn't guard well against 'noise' predictors. An expected result obtained from this study was the role played by sample size. On account of the larger number of available signals, a larger sample size often acted as a performance enhancer. The course of this study witnesses several instances where the larger samples improved both the accuracy and precision of methods including that for the traditional methods like Stepwise, Lasso etc.

Perhaps the most intriguing takeaway from this study was the overtly 'inclusive' nature of the stability mechanism. Different areas of the experiment often highlighted how this resulted in the stability selection consistently missing out some of the 'true' predictors and bringing in 'noise' predictors instead. Both these points seem to suggest that there is more work to be done as far as the accuracy of this method is concerned. The end of the experiment finally notes the overarching effect of normal random variates to disrupt the precision and accuracy of model selection methods. So far only incrementing the sample size used seems to be the legit defense against this impedance.

5.2 Future Work

As highlighted in section 5.1, one deviation observed for Stability selection method was its failure to augment Lasso in situations involving inherent variability of samples. This does seem like a counterintuitive suggestion, knowing that Stability selection has actually done comparatively better in the other scenarios highlighted within the study. Superficially, one might argue that the lambdas selected for the stability paths may not be rightly selected, however without a deeper analysis of the issue, one can't be certain of what's exactly at play. The experiment, however, did reveal one possible area of development for stability mechanism. The step involving the selection of a range of lambda values didn't play much of a meaningful role, considering that the lowest lambda was always bound to be selected; the lower lambda values actually consistently produced the highest selection count for predictors thus earning their place for being selected (as discussed in the pseudo code for stability selection in section 3.5). In that sense, it was inferred that this process, was actually redundant, offering no specific advantage and rather managed to increase the mean computational time of the overall mechanism. Another interesting revelation this study brought forward was an insight into the nature of 'noise' predictors. Categorically, not all 'noise' predictors used in the experimental study are truly 'noise'; some of them are 'noise' simply on the account of such predictors being correlated with the 'true' predictors via multicollinearity, as highlighted in section 3.1. Future research could extend the study conducted here by demarcating such true 'noise' predictors from correlated ones and conducting a finer examination on the behavior of such true 'noise' predictors. The tendency of the stability mechanism to include non-essential predictors in stochastic scenarios was observed at many instances in this study. While this could once more point out to the methodology used in selecting lambda parameters, increasing the lambdas beyond a point could actually lead to the exclusion of right predictors more often as well. One possible development, hence, could be to explore a method that helps choose a better, appropriate lambda parameter that considers this tradeoff more accurately. Adding on, inclusion of normal random variates often induced fluctuations that almost all model selection methods failed to handle. Considering the growth of high dimensional data problems, raising the immunity of the current model selection methods to such disruptive stochastic forces has to be studied further and developed on. Finally, it can be debated that using a larger sample can be a panacea for all the difficulties

mentioned in this section. However, considering that many areas of work still engaging with smaller population datasets, alternate strategies are certainly the need of the hour.

APPENDIX A
MONTE CARLO SIMULATION EXPERIMENT

The preliminary Monte Carlo (MC) simulation experiment is performed as follows:

1. Consider a simple regression model fitted with some predictors. The model is made to represent a population data generating process by assuming numerical ('population') values for the predictor coefficients.

   In this experiment, a simplified, one predictor, version of the logistic model formulated in equation 1.6 is implemented. The chosen predictor, student gender variable, has its factor levels assigned the following values,

   $$p_1 = 0.25 = \frac{\exp(\beta_0)}{1 + \exp(\beta_0)}$$, when the student gender variable is "Female"

   $$p_2 = 0.18 = \frac{\exp(\beta_0 + \beta_1 x_{1i})}{1 + \exp(\beta_0 + \beta_1 x_{1i})}$$, when the student gender variable is "Male"

   The resulting regression coefficients are β_0 = -1.0986 & β_1= -0.4177. Response values are then computed using an inverse logit of the linear predictor fitted with the above coefficient values.

2. The responses obtained in step 2 are randomized to obtain a response ('population') distribution. Samples are repeatedly drawn from this distribution and fitted with a regression model. The β's obtained at each draw are noted.

   This experiment drew samples of 500 observations, 1000 times from response distribution and fitted them with a logistic regression.

3. The average of the β's obtained over the all draws provides the MC simulated regression coefficient values.

The MC simulated regression coefficients were obtained as $\widehat{\beta_0}$ = -1.091985 & $\widehat{\beta_1}$ = -0.433672, close to the initially assumed ('population') parameter values. Statistical power of this test can be computed by the finding the average number of times the coefficients turned out statistically significant at the 95% confidence level. Attaching an indicator variable for the statistical significance of the β's at each draw and calculating their mean, essentially helps achieve this. The Likelihood Ratio Test used was thus found to have a power of 50.1%.

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